

Neural Dynamics of Directed and Random Exploration in an Immersive Decision-Making Task

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Introduction

- The Explore-exploit trade off is often split into **directed exploration** (seeking information; information bonus) and **random exploration** (decision noise)¹
- The Horizon Task isolates directed exploration from random exploration at the **first free choice**
- Directed exploration = information seeking in unequal trials (**information bonus**)
- Random exploration = choice of the poorer observed mean in equal trials (**decision noise**)
- Mid-frontal theta is a neural marker of uncertainty and prediction error shown to be involved in exploration²

1. Can we replicate previous results using the Horizon Task?
2. Can our seated VR-EEG setup yield usable frontocentral electrophysiological data in the Horizon task?

The Horizon Task¹

- Two-armed bandit task
- First four choices are forced with either **equal [2,2]** or **unequal [1,3]** information for the bandits
- Four forced choices are followed by either **one free decision (H1)** or **six free decisions (H6)**
- Rewards are normally distributed across bandits with reward differences of 4, 8, 12, and 20, and SD = 8



Methods - Setup

Participants & Procedure

- N = 4 (Preliminary/ Pilot data)
- 4 practice games + 96 games per participant

Apparatus

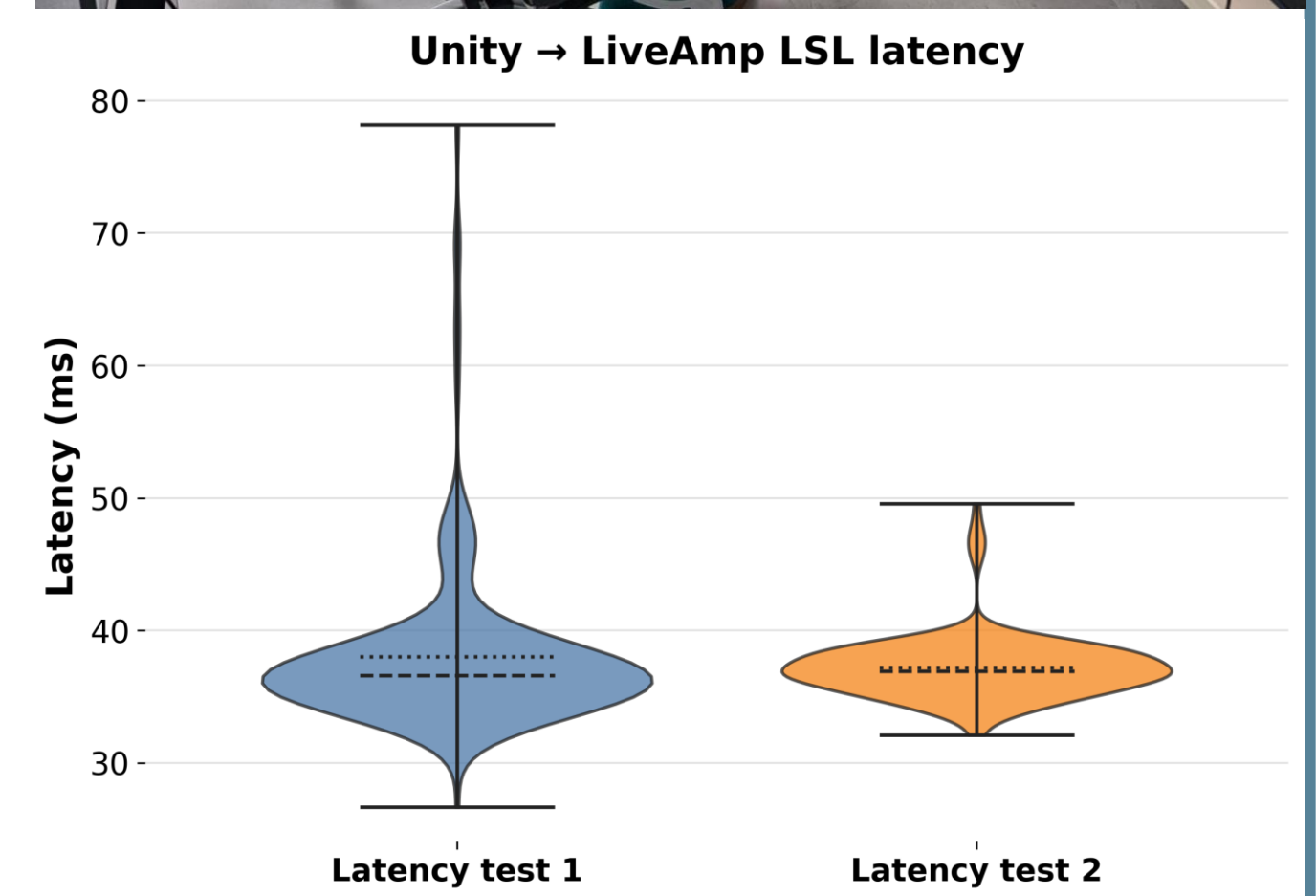
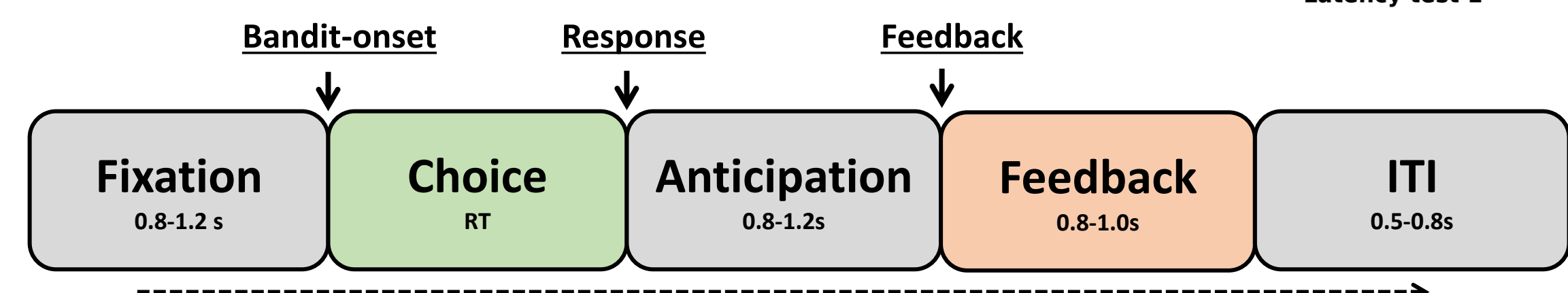
- HTC Vive Focus 3 (SteamVR, 90Hz) with eye-tracking recorded at 90 Hz
- Unity environment: Horizon casino scene
- Brain Products LiveAmp 64 (wireless, 64 electrodes + REF + GND)



Synchronisation³

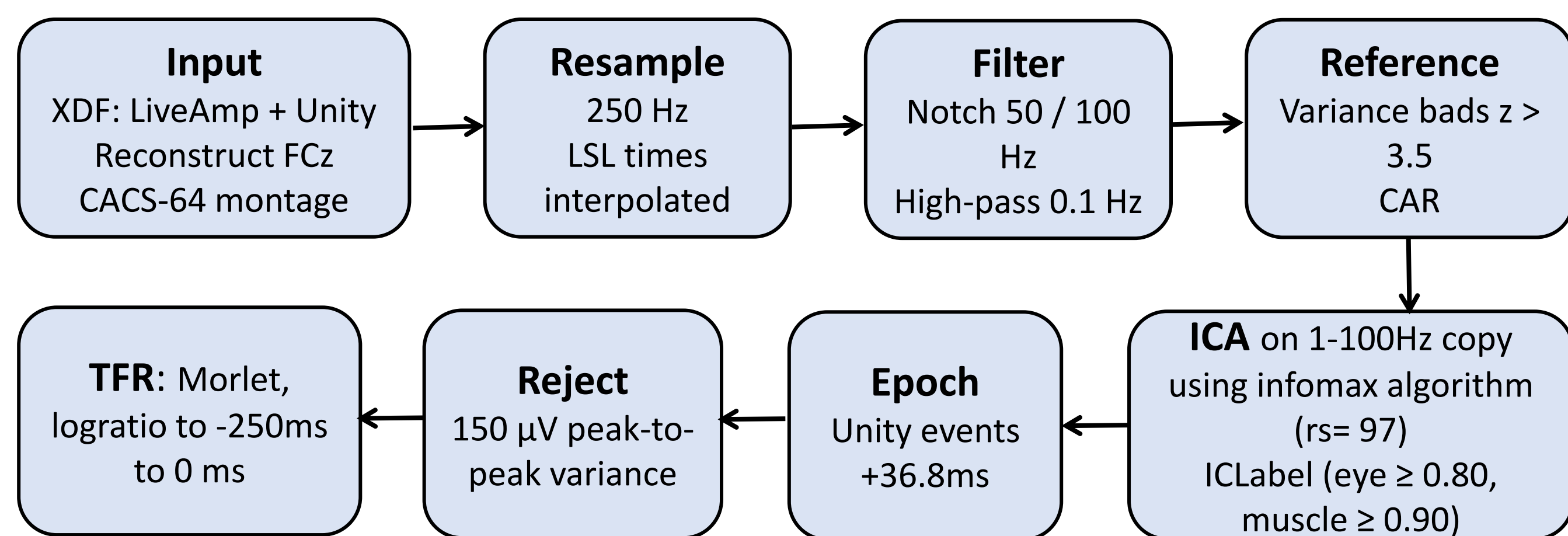
- LSL4Unity & LiveAmp LSL Connector
- Recorded via LabRecorder
- Offline Latency check: photodiode in HMD + StimTrak vs Unity event markers

Trial Structure

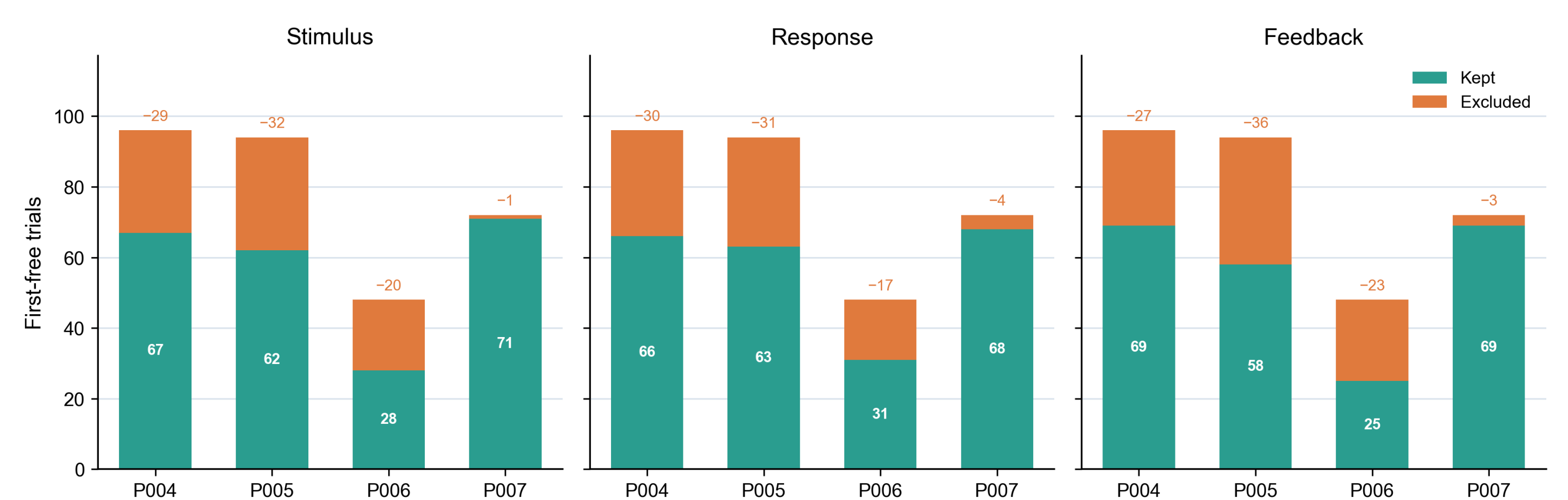


Electrophysiological Data Preprocessing

Data Preprocessing Pipeline



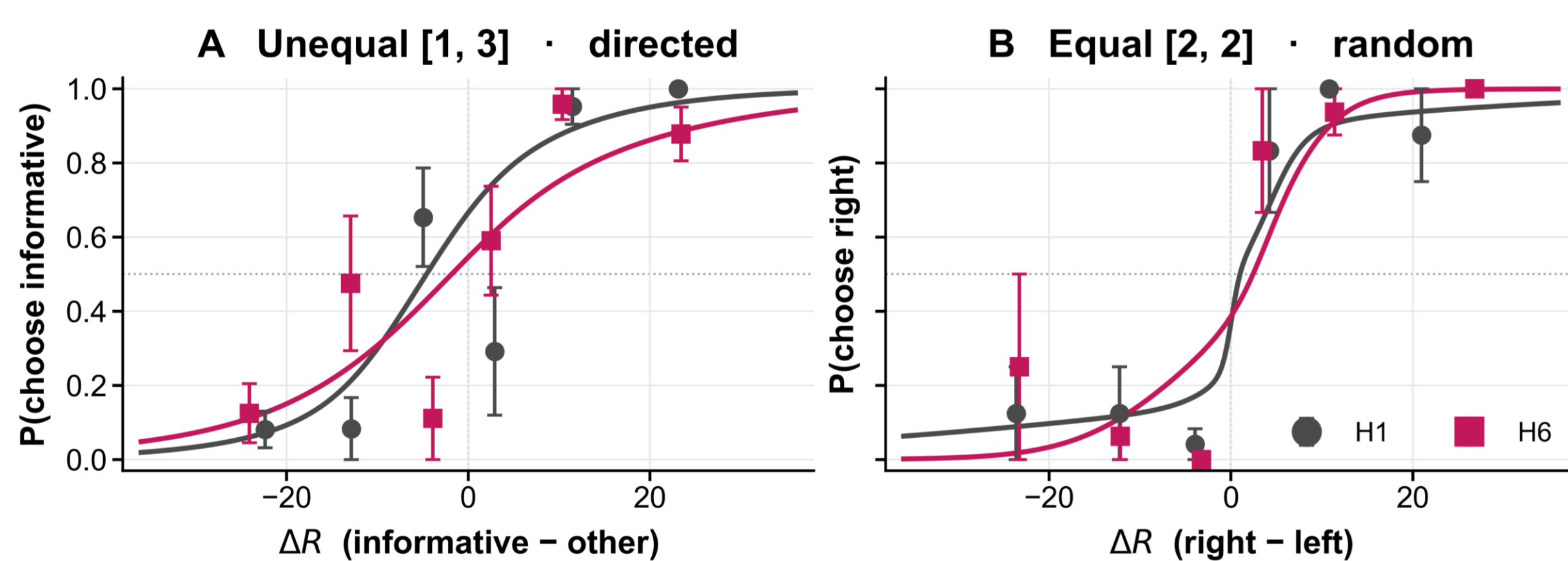
First-free trials before and after peak-to-peak variance rejection



Behavioral Data

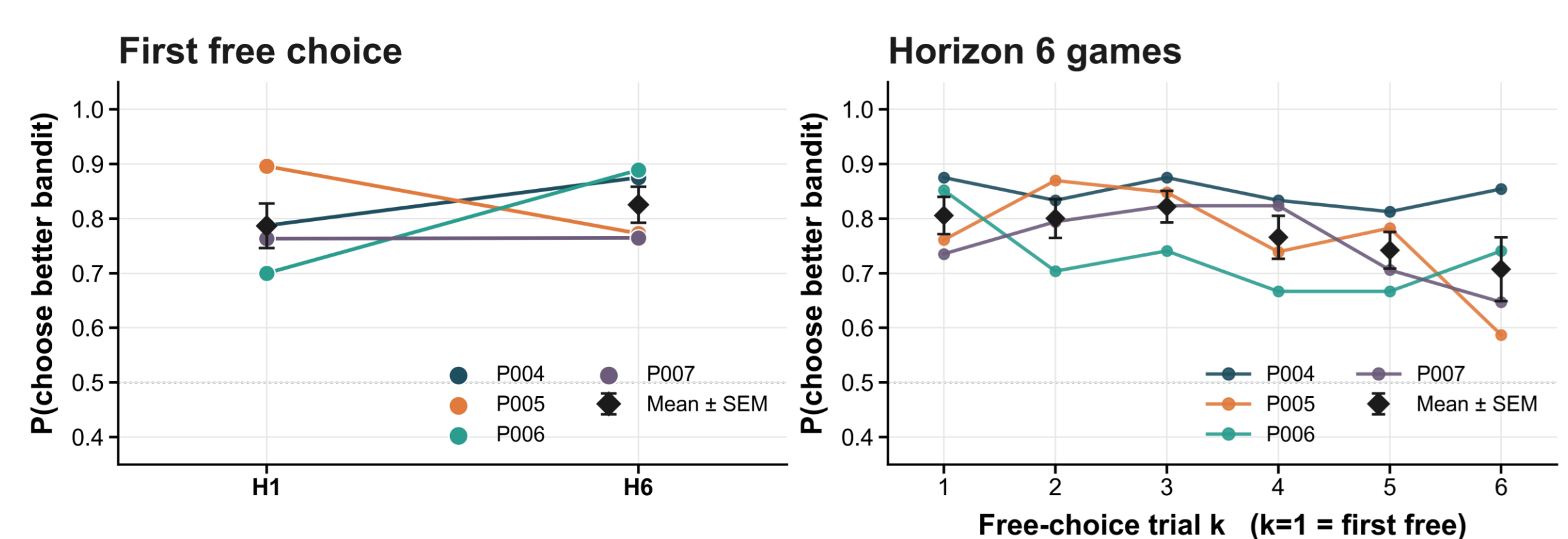
1. Can we replicate previous results using the Horizon Task?

Choice Curves – First-free choice



- How often people pick the informative arm [1,3] or the right arm [2,2] as a function of the observed reward difference ΔR
- Wilson et al. (2014): in H6 the [1,3] curve should shift toward the informative option (directed exploration), and the [2,2] curve should get shallower (random exploration).
- Here both horizons look mainly greedy, meaning there is no stable H6 shift or slope change

Task Performance

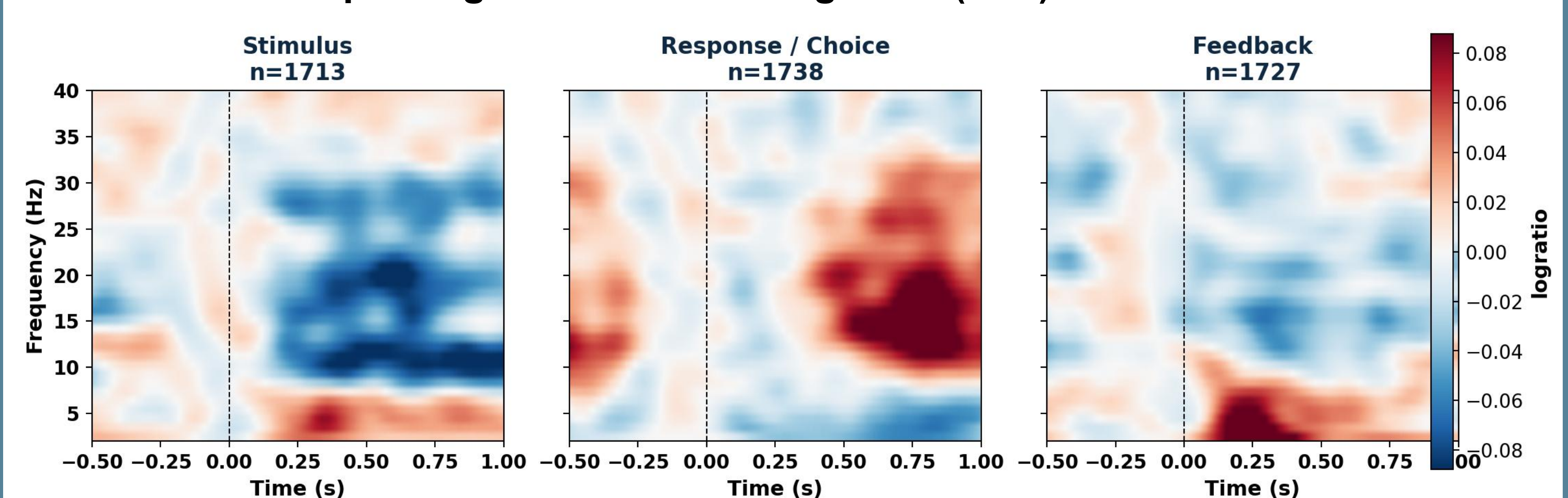


- Wilson et al. (2014): people should be well above chance and get better within H6 games as they use the extra samples
- Here performance is clearly above chance, but there is no clean H6 learning rise

Electrophysiological Data (Exploratory)

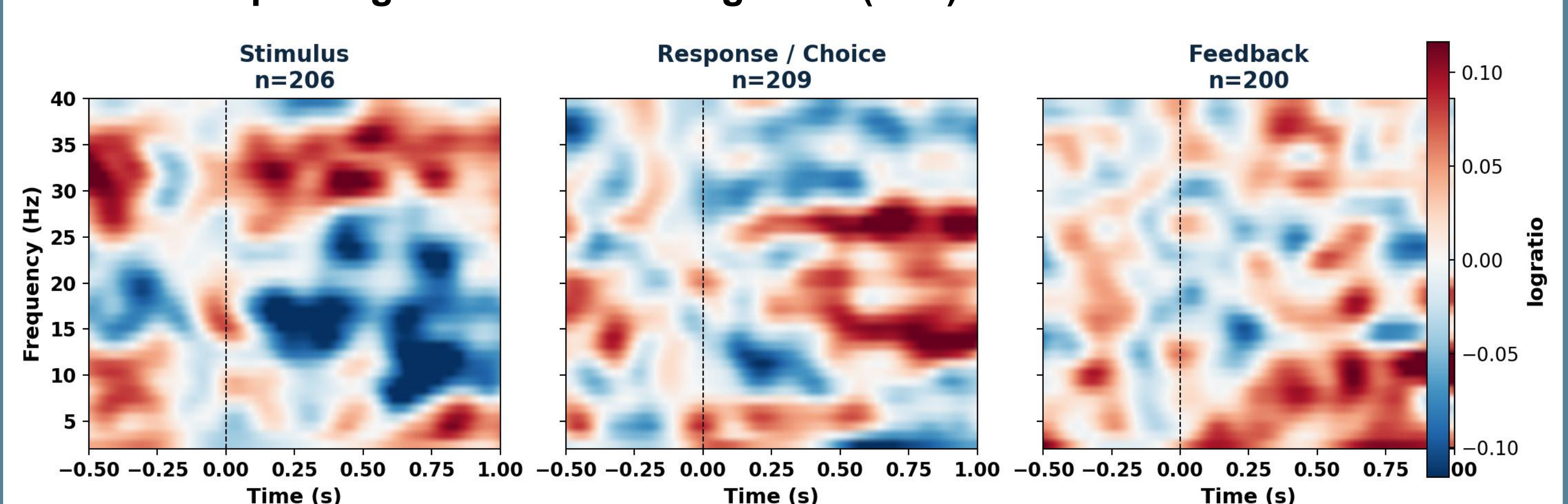
2. Can we get usable electrophysiological data from our setup?

Spectrogram: Grand Average FCz (N=4) - All trials



- Stimulus (Bandit active):** after onset, power drops around 10-30 Hz (alpha/beta desynchronization) and some increase below ~8 Hz
- Response:** beta increase (~10-25 Hz, ~0.4-1.0 s); Potentially, a post-movement pattern after a button press
- Feedback:** a brief low-frequency (theta) increase just after reward then alpha/beta decrease

Spectrogram: Grand Average FCz (N=4) - First-free choice



References

1. Wilson, R. C., Geana, A., White, J. M., Ludvig, E. A., & Cohen, J. D. (2014). Humans use directed and random exploration to solve the explore-exploit dilemma. *Journal of Experimental Psychology: General*, 143(6), 2074–2081. <https://doi.org/10.1037/a0038199>
2. Cavanagh, J. F., Figueroa, C. M., Cohen, M. X., & Frank, M. J. (2012). Frontal theta reflects uncertainty and unexpectedness during exploration and exploitation. *Cerebral Cortex*, 22(11), 2575–2586. <https://doi.org/10.1093/cercor/bhr332>
3. Krampe, C., Wolfers, J., & Dean, P. (2025). The integration of virtual reality and EEG: A step-by-step guideline. *MethodsX*, 16, Article 103770. <https://doi.org/10.1016/j.mex.2025.103770>

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